

MODEL TRAINING REPORT

DGA Gas Composition - Virtual GC (multi-label, 5 gas)

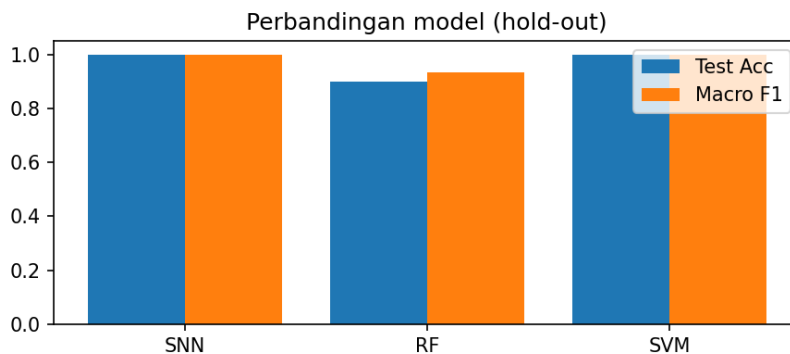
Models: SNN (LIF), Random Forest, SVM

Date: 2026-10-07 14:11:03

1. Training Summary (hold-out validation)

Model	Train Acc	Test Acc	Macro F1	Hamming	Train Time (s)	Inference (ms)
SNN	1.0000	1.0000	1.0000	0.0000	27.14	4.095
RF	1.0000	0.9000	0.9333	0.0200	23.68	5.291
SVM	1.0000	1.0000	1.0000	0.0000	0.71	0.999

Acc = subset accuracy (semua 5 label harus benar). Validasi hold-out hanya 10 sampel, sehingga satu salah tebak mengubah skor cukup besar. Gunakan LOMO dan K-Fold sebagai ukuran yang lebih stabil.



2. Model Configuration

Task	Multi-label gas composition, keluaran confidence 0-100% per gas: AIR, C2H2, C2H4, H2, CH4
Samples (asli)	50 sampel; hold-out stratified: 40 latih / 10 validasi
Feature Extraction	TSFRESH EfficientFCParameters, 34 sensor x 288 timestep
Feature Dimensions	7530 fitur (union seleksi fitur per gas), StandardScaler
Validation Method	Hold-out stratified 80/20, K-Fold (5), LOMO (1 file per fold), LOCO
SNN	FC 7530 - 128 - 64 - 5, neuron LIF (beta 0.95, 25 time step, rate coding), BCE loss, Adam lr 1e-3, early stopping (patience 15). Parameter: 972,549. File: 19187.7 KB
Random Forest	n_estimators=200, max_depth=10. File: 1153.3 KB
SVM	kernel=rbf, C=1, probability=True. File: 8588.2 KB
Inference Latency / sample	SNN 4.095 ms, RF 5.291 ms, SVM 0.999 ms
Device	Tesla T4
Peak RAM (proses)	1655 MB
Total Pipeline Time	757.1 s

3. Sensors & Power Consumption

Kanal data yang dipakai (34 kanal):

mos0_mV	mos1_mV	mos2_mV	mos3_mV
mos4_mV	mos5_mV	mos6_mV	mos7_mV
mos8_mV	mos9_mV	mos10_mV	mos11_mV
mos12_mV	mos13_mV	mos14_mV	mos15_mV

temp_chamber_C	temp_oil_C	humidity_chamber_pct	humidity_oil_pct
ina226_1_vbus_V	ina226_1_current_mA	ina226_1_power_W	ina226_2_vbus_V
ina226_2_current_mA	ina226_2_power_W	kria_vbus_V	kria_current_mA
kria_power_W	ndir_ratio	ndir_baseline	ndir_response_pct
ndir_absorbance_au	flow_inlet		

Daya perangkat saat akuisisi data (dari kolom data mentah; bukan daya saat training):

Kanal	Rata-rata (W)	Maks (W)
ina226_1_power_W	13.875	21.034
ina226_2_power_W	7.049	8.408
kria_power_W	3.276	4.920

4. K-Fold Cross Validation (data asli)

Model	Folds	Acc (mean)	Acc (std)	F1 (mean)	F1 (std)
Random Forest	5	0.8800	0.1166	0.8307	0.1080
SVM	5	0.8400	0.0490	0.7773	0.0666

5. Leave-One-Measurement-Out (LOMO) - Random Forest

Setiap fold menguji 1 file dan melatih dengan file lainnya. Total fold: 50.

Metric	Estimate	95% CI (lower)	95% CI (upper)
Subset Accuracy	0.8600	0.7600	0.9600
Macro F1	0.9214	0.8371	0.9750
Hamming Loss	0.0280	-	-

CI: bootstrap nonparametrik (1000 resample, percentile 2.5-97.5) atas 50 prediksi.

Gas	Precision	Recall	F1-Score	Support
AIR	1.0000	0.8000	0.8889	10
C2H2	1.0000	0.7000	0.8235	10
C2H4	1.0000	0.9000	0.9474	10
H2	1.0000	0.9000	0.9474	10
CH4	1.0000	1.0000	1.0000	10
Macro avg	1.0000	0.8600	0.9214	50

Confusion matrix (prediksi = gas dengan confidence tertinggi)

Aktual / Prediksi	AIR	C2H2	C2H4	H2	CH4
AIR	10	0	0	0	0
C2H2	0	10	0	0	0
C2H4	0	0	10	0	0
H2	0	0	0	10	0
CH4	0	0	0	0	10

Hasil per file (satu fold = satu file)

File	Aktual	Prediksi	Hasil	Conf. gas aktual
AIR/DGA_PFA10_Baseline_2026-09-24T17-40-54.csv	AIR	-	SALAH	43.0%
AIR/DGA_PFA1_Baseline_2026-09-24T16-44-46.csv	AIR	AIR	OK	53.0%
AIR/DGA_PFA2_Baseline_2026-09-24T16-50-06.csv	AIR	AIR	OK	79.0%
AIR/DGA_PFA3_Baseline_2026-09-24T16-55-19.csv	AIR	AIR	OK	84.0%
AIR/DGA_PFA4_Baseline_2026-09-24T17-00-34.csv	AIR	AIR	OK	86.0%
AIR/DGA_PFA5_Baseline_2026-09-24T17-06-38.csv	AIR	AIR	OK	90.0%
AIR/DGA_PFA6_Baseline_2026-09-24T17-12-52.csv	AIR	AIR	OK	88.0%
AIR/DGA_PFA7_Baseline_2026-09-24T17-19-03.csv	AIR	AIR	OK	80.0%
AIR/DGA_PFA8_Baseline_2026-09-24T17-30-00.csv	AIR	AIR	OK	73.0%
AIR/DGA_PFA9_Baseline_2026-09-24T17-35-36.csv	AIR	-	SALAH	45.0%
C2H2/DGA_PFAAsethilena10_Baseline_2026-09-25T14-21-1	C2H2	C2H2	OK	56.0%
C2H2/DGA_PFAAsethilena1_Baseline_2026-09-25T13-26-29	C2H2	-	SALAH	44.0%
C2H2/DGA_PFAAsethilena2_Baseline_2026-09-25T13-32-49	C2H2	C2H2	OK	56.0%
C2H2/DGA_PFAAsethilena3_Baseline_2026-09-25T13-38-30	C2H2	-	SALAH	49.0%
C2H2/DGA_PFAAsethilena4_Baseline_2026-09-25T13-43-59	C2H2	-	SALAH	39.0%
C2H2/DGA_PFAAsethilena5_Baseline_2026-09-25T13-53-50	C2H2	C2H2	OK	76.0%
C2H2/DGA_PFAAsethilena6_Baseline_2026-09-25T14-00-27	C2H2	C2H2	OK	58.0%

File	Aktual	Prediksi	Hasil	Conf. gas aktual
C2H2/DGA_PFAAsethilen7_Baseline_2026-09-25T14-05-41	C2H2	C2H2	OK	73.0%
C2H2/DGA_PFAAsethilen8_Baseline_2026-09-25T14-10-54	C2H2	C2H2	OK	74.0%
C2H2/DGA_PFAAsethilen9_Baseline_2026-09-25T14-16-07	C2H2	C2H2	OK	68.0%
C2H4/DGA_PFAET10_Baseline_2026-09-24T18-43-15.csv	C2H4	C2H4	OK	88.0%
C2H4/DGA_PFAET1_Baseline_2026-09-24T17-47-20.csv	C2H4	-	SALAH	45.0%
C2H4/DGA_PFAET2_Baseline_2026-09-24T17-53-42.csv	C2H4	C2H4	OK	76.0%
C2H4/DGA_PFAET3_Baseline_2026-09-24T18-01-07.csv	C2H4	C2H4	OK	93.0%
C2H4/DGA_PFAET4_Baseline_2026-09-24T18-07-06.csv	C2H4	C2H4	OK	98.0%
C2H4/DGA_PFAET5_Baseline_2026-09-24T18-14-48.csv	C2H4	C2H4	OK	99.0%
C2H4/DGA_PFAET6_Baseline_2026-09-24T18-20-50.csv	C2H4	C2H4	OK	95.0%
C2H4/DGA_PFAET7_Baseline_2026-09-24T18-26-03.csv	C2H4	C2H4	OK	90.0%
C2H4/DGA_PFAET8_Baseline_2026-09-24T18-31-49.csv	C2H4	C2H4	OK	94.0%
C2H4/DGA_PFAET9_Baseline_2026-09-24T18-37-40.csv	C2H4	C2H4	OK	96.0%
CH4/DGA_PFAASwap10_Baseline_2026-09-25T13-19-27.csv	CH4	CH4	OK	63.0%
CH4/DGA_PFAASwap1_Baseline_2026-09-25T12-19-29.csv	CH4	CH4	OK	84.0%
CH4/DGA_PFAASwap2_Baseline_2026-09-25T12-26-38.csv	CH4	CH4	OK	95.0%
CH4/DGA_PFAASwap3_Baseline_2026-09-25T12-33-57.csv	CH4	CH4	OK	100.0%
CH4/DGA_PFAASwap4_Baseline_2026-09-25T12-39-14.csv	CH4	CH4	OK	97.0%
CH4/DGA_PFAASwap5_Baseline_2026-09-25T12-45-58.csv	CH4	CH4	OK	100.0%
CH4/DGA_PFAASwap6_Baseline_2026-09-25T12-53-06.csv	CH4	CH4	OK	98.0%
CH4/DGA_PFAASwap7_Baseline_2026-09-25T12-59-38.csv	CH4	CH4	OK	97.0%
CH4/DGA_PFAASwap8_Baseline_2026-09-25T13-08-21.csv	CH4	CH4	OK	99.0%
CH4/DGA_PFAASwap9_Baseline_2026-09-25T13-14-10.csv	CH4	CH4	OK	92.0%
H2/DGA_H2_10_Baseline_2026-09-28T07-23-12.csv	H2	H2	OK	99.0%
H2/DGA_H2_1_Baseline_2026-09-28T06-02-24.csv	H2	H2	OK	85.0%
H2/DGA_H2_2_Baseline_2026-09-28T06-08-25.csv	H2	H2	OK	87.0%
H2/DGA_H2_3_Baseline_2026-09-28T06-14-25.csv	H2	H2	OK	97.0%
H2/DGA_H2_4_Baseline_2026-09-28T06-20-23.csv	H2	H2	OK	96.0%
H2/DGA_H2_5_Baseline_2026-09-28T06-26-36.csv	H2	H2	OK	93.0%
H2/DGA_H2_6_Baseline_2026-09-28T06-32-25.csv	H2	H2	OK	91.0%
H2/DGA_H2_7_Baseline_2026-09-28T06-38-25.csv	H2	H2	OK	100.0%
H2/DGA_H2_8_Baseline_2026-09-28T06-46-05.csv	H2	H2	OK	83.0%
H2/DGA_H2_9_Baseline_2026-09-28T06-51-16.csv	H2	-	SALAH	34.0%

5. Leave-One-Measurement-Out (LOMO) - SVM

Setiap fold menguji 1 file dan melatih dengan file lainnya. Total fold: 50.

Metric	Estimate	95% CI (lower)	95% CI (upper)
Subset Accuracy	0.8400	0.7400	0.9400
Macro F1	0.9109	0.8256	0.9666
Hamming Loss	0.0320	-	-

CI: bootstrap nonparametrik (1000 resample, percentile 2.5-97.5) atas 50 prediksi.

Gas	Precision	Recall	F1-Score	Support
AIR	1.0000	0.8000	0.8889	10
C2H2	1.0000	0.9000	0.9474	10
C2H4	1.0000	0.9000	0.9474	10
H2	1.0000	0.7000	0.8235	10
CH4	1.0000	0.9000	0.9474	10
Macro avg	1.0000	0.8400	0.9109	50

Confusion matrix (prediksi = gas dengan confidence tertinggi)

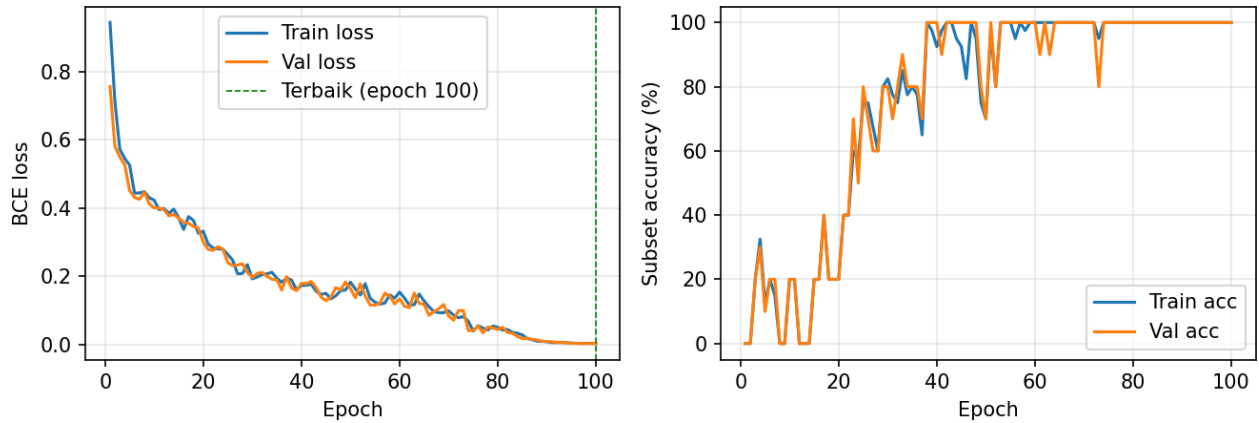
Aktual / Prediksi	AIR	C2H2	C2H4	H2	CH4
AIR	9	0	0	0	1
C2H2	1	9	0	0	0
C2H4	0	0	9	0	1
H2	0	0	0	7	3
CH4	1	0	0	0	9

Hasil per file (satu fold = satu file)

File	Aktual	Prediksi	Hasil	Conf. gas aktual
AIR/DGA_PFA10_Baseline_2026-09-24T17-40-54.csv	AIR	AIR	OK	92.9%
AIR/DGA_PFA1_Baseline_2026-09-24T16-44-46.csv	AIR	-	SALAH	6.3%
AIR/DGA_PFA2_Baseline_2026-09-24T16-50-06.csv	AIR	AIR	OK	70.6%
AIR/DGA_PFA3_Baseline_2026-09-24T16-55-19.csv	AIR	AIR	OK	98.9%
AIR/DGA_PFA4_Baseline_2026-09-24T17-00-34.csv	AIR	AIR	OK	96.7%
AIR/DGA_PFA5_Baseline_2026-09-24T17-06-38.csv	AIR	AIR	OK	98.7%
AIR/DGA_PFA6_Baseline_2026-09-24T17-12-52.csv	AIR	AIR	OK	98.6%
AIR/DGA_PFA7_Baseline_2026-09-24T17-19-03.csv	AIR	AIR	OK	95.3%
AIR/DGA_PFA8_Baseline_2026-09-24T17-30-00.csv	AIR	AIR	OK	98.0%
AIR/DGA_PFA9_Baseline_2026-09-24T17-35-36.csv	AIR	-	SALAH	47.4%
C2H2/DGA_PFAAsethilena10_Baseline_2026-09-25T14-21-1	C2H2	C2H2	OK	64.4%
C2H2/DGA_PFAAsethilena1_Baseline_2026-09-25T13-26-29	C2H2	-	SALAH	7.0%
C2H2/DGA_PFAAsethilena2_Baseline_2026-09-25T13-32-49	C2H2	C2H2	OK	74.2%
C2H2/DGA_PFAAsethilena3_Baseline_2026-09-25T13-38-30	C2H2	C2H2	OK	97.1%
C2H2/DGA_PFAAsethilena4_Baseline_2026-09-25T13-43-59	C2H2	C2H2	OK	92.9%
C2H2/DGA_PFAAsethilena5_Baseline_2026-09-25T13-53-50	C2H2	C2H2	OK	97.6%
C2H2/DGA_PFAAsethilena6_Baseline_2026-09-25T14-00-27	C2H2	C2H2	OK	93.9%

File	Aktual	Prediksi	Hasil	Conf. gas aktual
C2H2/DGA_PFAAsethilena7_Baseline_2026-09-25T14-05-41	C2H2	C2H2	OK	97.3%
C2H2/DGA_PFAAsethilena8_Baseline_2026-09-25T14-10-54	C2H2	C2H2	OK	94.5%
C2H2/DGA_PFAAsethilena9_Baseline_2026-09-25T14-16-07	C2H2	C2H2	OK	96.6%
C2H4/DGA_PFAET10_Baseline_2026-09-24T18-43-15.csv	C2H4	C2H4	OK	90.3%
C2H4/DGA_PFAET1_Baseline_2026-09-24T17-47-20.csv	C2H4	C2H4	OK	73.8%
C2H4/DGA_PFAET2_Baseline_2026-09-24T17-53-42.csv	C2H4	C2H4	OK	89.0%
C2H4/DGA_PFAET3_Baseline_2026-09-24T18-01-07.csv	C2H4	C2H4	OK	93.7%
C2H4/DGA_PFAET4_Baseline_2026-09-24T18-07-06.csv	C2H4	-	SALAH	8.2%
C2H4/DGA_PFAET5_Baseline_2026-09-24T18-14-48.csv	C2H4	C2H4	OK	98.6%
C2H4/DGA_PFAET6_Baseline_2026-09-24T18-20-50.csv	C2H4	C2H4	OK	96.6%
C2H4/DGA_PFAET7_Baseline_2026-09-24T18-26-03.csv	C2H4	C2H4	OK	56.1%
C2H4/DGA_PFAET8_Baseline_2026-09-24T18-31-49.csv	C2H4	C2H4	OK	97.0%
C2H4/DGA_PFAET9_Baseline_2026-09-24T18-37-40.csv	C2H4	C2H4	OK	98.2%
CH4/DGA_PFAASwap10_Baseline_2026-09-25T13-19-27.csv	CH4	CH4	OK	82.3%
CH4/DGA_PFAASwap1_Baseline_2026-09-25T12-19-29.csv	CH4	-	SALAH	12.6%
CH4/DGA_PFAASwap2_Baseline_2026-09-25T12-26-38.csv	CH4	CH4	OK	95.4%
CH4/DGA_PFAASwap3_Baseline_2026-09-25T12-33-57.csv	CH4	CH4	OK	96.6%
CH4/DGA_PFAASwap4_Baseline_2026-09-25T12-39-14.csv	CH4	CH4	OK	96.4%
CH4/DGA_PFAASwap5_Baseline_2026-09-25T12-45-58.csv	CH4	CH4	OK	98.5%
CH4/DGA_PFAASwap6_Baseline_2026-09-25T12-53-06.csv	CH4	CH4	OK	98.2%
CH4/DGA_PFAASwap7_Baseline_2026-09-25T12-59-38.csv	CH4	CH4	OK	97.6%
CH4/DGA_PFAASwap8_Baseline_2026-09-25T13-08-21.csv	CH4	CH4	OK	77.1%
CH4/DGA_PFAASwap9_Baseline_2026-09-25T13-14-10.csv	CH4	CH4	OK	97.6%
H2/DGA_H2_10_Baseline_2026-09-28T07-23-12.csv	H2	H2	OK	85.0%
H2/DGA_H2_1_Baseline_2026-09-28T06-02-24.csv	H2	H2	OK	70.6%
H2/DGA_H2_2_Baseline_2026-09-28T06-08-25.csv	H2	-	SALAH	7.8%
H2/DGA_H2_3_Baseline_2026-09-28T06-14-25.csv	H2	H2	OK	96.5%
H2/DGA_H2_4_Baseline_2026-09-28T06-20-23.csv	H2	H2	OK	94.9%
H2/DGA_H2_5_Baseline_2026-09-28T06-26-36.csv	H2	H2	OK	95.5%
H2/DGA_H2_6_Baseline_2026-09-28T06-32-25.csv	H2	H2	OK	96.8%
H2/DGA_H2_7_Baseline_2026-09-28T06-38-25.csv	H2	H2	OK	89.4%
H2/DGA_H2_8_Baseline_2026-09-28T06-46-05.csv	H2	-	SALAH	7.7%
H2/DGA_H2_9_Baseline_2026-09-28T06-51-16.csv	H2	-	SALAH	8.7%

6. SNN - Riwayat Training per Epoch



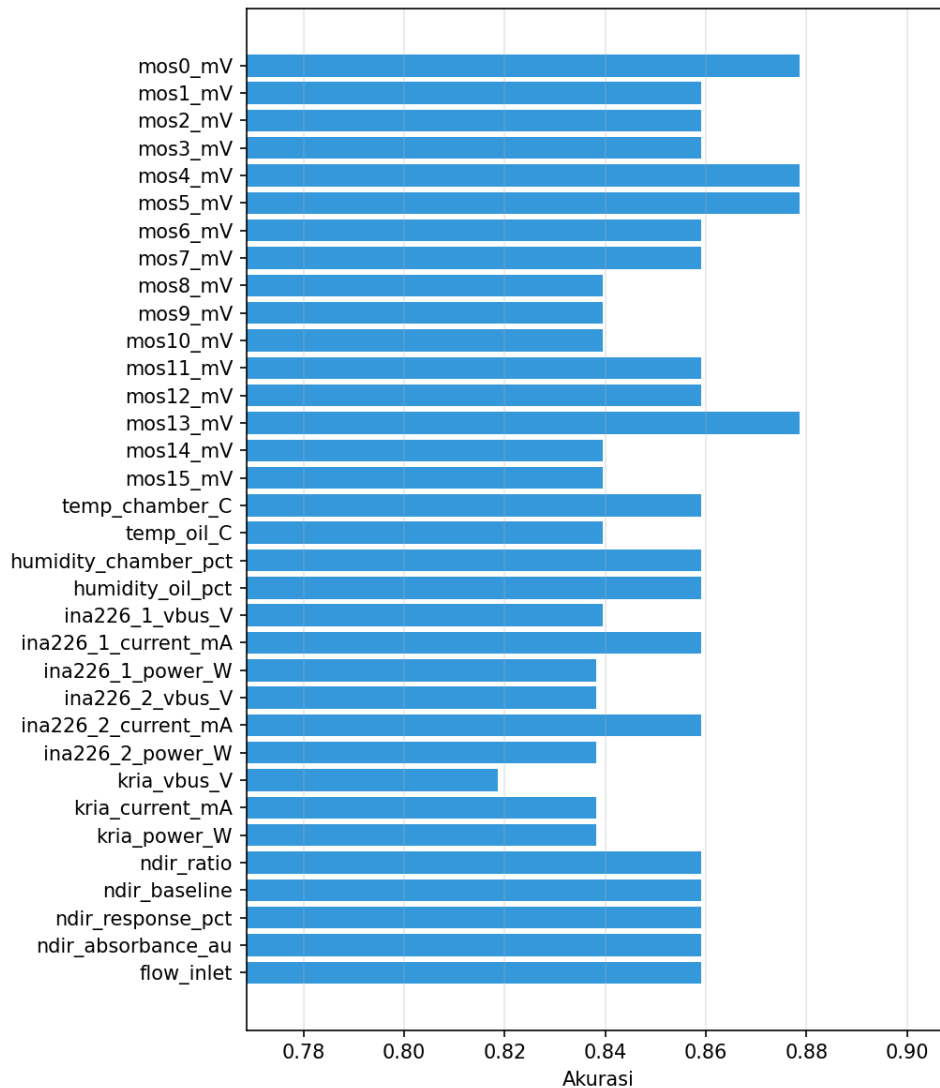
Total epoch berjalan: 100. Epoch terbaik (val loss terendah): 100, val loss 0.0021. Model yang disimpan adalah dari epoch terbaik.

Epoch	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
1	0.00	0.00	0.9447	0.7566
2	0.00	0.00	0.7122	0.5811
3	20.00	20.00	0.5729	0.5497
4	32.50	30.00	0.5454	0.5274
5	12.50	10.00	0.5261	0.4516
6	20.00	20.00	0.4440	0.4311
7	15.00	20.00	0.4441	0.4258
8	0.00	0.00	0.4481	0.4470
9	0.00	0.00	0.4301	0.4123
10	20.00	20.00	0.4231	0.4001
11	20.00	20.00	0.3953	0.4002
12	0.00	0.00	0.3988	0.3952
13	0.00	0.00	0.3846	0.3771
14	0.00	0.00	0.3966	0.3815
15	20.00	20.00	0.3716	0.3696
16	20.00	20.00	0.3370	0.3585
17	40.00	40.00	0.3747	0.3551
18	20.00	20.00	0.3636	0.3459
19	20.00	20.00	0.3261	0.3410
20	20.00	20.00	0.3319	0.3003
21	40.00	40.00	0.2944	0.2786
22	40.00	40.00	0.2828	0.2756
23	60.00	70.00	0.2798	0.2866
24	57.50	50.00	0.2794	0.2792
25	75.00	80.00	0.2637	0.2396
26	75.00	70.00	0.2466	0.2312
27	67.50	60.00	0.2070	0.2324

Epoch	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
28	60.00	60.00	0.2079	0.2366
29	80.00	80.00	0.2338	0.2106
30	82.50	80.00	0.1921	0.1963
31	77.50	70.00	0.1975	0.2080
32	75.00	80.00	0.2046	0.2107
33	85.00	90.00	0.2070	0.1995
34	77.50	80.00	0.2114	0.1908
35	80.00	80.00	0.1941	0.1890
36	77.50	80.00	0.1819	0.1591
37	65.00	70.00	0.1944	0.1986
38	100.00	100.00	0.1885	0.1651
39	97.50	100.00	0.1600	0.1577
40	92.50	100.00	0.1724	0.1784
41	97.50	90.00	0.1733	0.1785
42	100.00	100.00	0.1762	0.1842
43	100.00	100.00	0.1559	0.1664
44	95.00	100.00	0.1464	0.1388
45	92.50	100.00	0.1495	0.1275
46	82.50	100.00	0.1333	0.1382
47	100.00	100.00	0.1421	0.1661
48	95.00	100.00	0.1575	0.1597
49	75.00	80.00	0.1593	0.1829
50	70.00	70.00	0.1824	0.1632
51	97.50	100.00	0.1622	0.1364
52	80.00	80.00	0.1446	0.1774
53	100.00	100.00	0.1779	0.1418
54	100.00	100.00	0.1351	0.1154
55	100.00	100.00	0.1236	0.1144
56	95.00	100.00	0.1177	0.1207
57	100.00	100.00	0.1211	0.1497
58	97.50	100.00	0.1449	0.1416
59	100.00	100.00	0.1351	0.1183
60	100.00	100.00	0.1532	0.1332
61	100.00	90.00	0.1358	0.1118
62	100.00	100.00	0.1148	0.1073
63	100.00	90.00	0.1161	0.1504
64	100.00	100.00	0.1476	0.1202
65	100.00	100.00	0.1268	0.1172
66	100.00	100.00	0.1100	0.0849
67	100.00	100.00	0.0957	0.0957

Epoch	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
68	100.00	100.00	0.0926	0.1044
69	100.00	100.00	0.0927	0.1166
70	100.00	100.00	0.0982	0.0829
71	100.00	100.00	0.0855	0.0707
72	100.00	100.00	0.0779	0.0989
73	95.00	80.00	0.0814	0.0990
74	100.00	100.00	0.0679	0.0403
75	100.00	100.00	0.0389	0.0401
76	100.00	100.00	0.0545	0.0534
77	100.00	100.00	0.0477	0.0332
78	100.00	100.00	0.0420	0.0507
79	100.00	100.00	0.0533	0.0477
80	100.00	100.00	0.0498	0.0435
81	100.00	100.00	0.0413	0.0499
82	100.00	100.00	0.0433	0.0346
83	100.00	100.00	0.0348	0.0324
84	100.00	100.00	0.0324	0.0232
85	100.00	100.00	0.0275	0.0165
86	100.00	100.00	0.0168	0.0178
87	100.00	100.00	0.0133	0.0136
88	100.00	100.00	0.0091	0.0130
89	100.00	100.00	0.0082	0.0090
90	100.00	100.00	0.0075	0.0079
91	100.00	100.00	0.0050	0.0074
92	100.00	100.00	0.0049	0.0054
93	100.00	100.00	0.0051	0.0056
94	100.00	100.00	0.0044	0.0040
95	100.00	100.00	0.0032	0.0033
96	100.00	100.00	0.0028	0.0026
97	100.00	100.00	0.0026	0.0023
98	100.00	100.00	0.0023	0.0024
99	100.00	100.00	0.0025	0.0023
100	100.00	100.00	0.0022	0.0021

7. LOCO - Akurasi tanpa tiap sensor (Random Forest)



Selisih kecil antar sensor masih dalam rentang noise untuk jumlah sampel yang sedikit.

8. Catatan Pembacaan Hasil

- Dataset hanya gas murni dengan sedikit sampel per gas; hasil belum mewakili campuran gas minyak trafo.
- Hold-out hanya berisi beberapa sampel sehingga skor 1.00 belum membuktikan model sempurna.
- LOMO menguji file yang belum pernah dilihat, tetapi file dari sesi rekaman yang sama masih ada di data latih.
- LOMO hanya dijalankan untuk RF dan SVM; SNN dievaluasi pada hold-out.

9. Log Lengkap Training

MENGINSTALL SNN TORCH...

DGA_ML GAS COMPOSITION PIPELINE
Multi-Label Prediction + IEC 60599 Fault Diagnosis
Models: SNN (LIF) | Random Forest | SVM

[0/8] Loading Dataset...

Loaded 50 samples dari 5 gas classes
AIR: 10 samples
C2H2: 10 samples
C2H4: 10 samples
H2: 10 samples
CH4: 10 samples

[1/8] Extracting TSFRESH Features...

Terekstrak 26418 fitur raw.
Memilih fitur relevan per gas (union selection)...
AIR: 995 fitur relevan
C2H2: 0 fitur relevan
C2H4: 2049 fitur relevan
H2: 1399 fitur relevan
CH4: 4790 fitur relevan
Total fitur setelah union selection: 7530

[3/8] Training Multi-Output Models...

[1/3] Training Spiking Neural Network (LIF)...

Perangkat: cuda
Epoch 20/100 Train Loss: 0.3319, Val Loss: 0.3003
Epoch 40/100 Train Loss: 0.1724, Val Loss: 0.1784
Epoch 60/100 Train Loss: 0.1532, Val Loss: 0.1332
Epoch 80/100 Train Loss: 0.0498, Val Loss: 0.0435
Epoch 100/100 Train Loss: 0.0022, Val Loss: 0.0021
SNN Subset Acc: 1.0000 | Macro F1: 1.0000 | Hamming Loss: 0.0000 | Waktu: 27.14s | Inferensi: 4.09ms

[2/3] Training Random Forest (Multi-Output)...

RF Best Params: {'estimator__max_depth': 10, 'estimator__n_estimators': 200}
RF Subset Acc: 0.9000 | Macro F1: 0.9333 | Hamming Loss: 0.0200 | Waktu: 23.68s | Inferensi: 5.29ms

[3/3] Training SVM (Multi-Output, RBF)...

SVM Best Params: {'estimator__C': 1, 'estimator__kernel': 'rbf'}
SVM Subset Acc: 1.0000 | Macro F1: 1.0000 | Hamming Loss: 0.0000 | Waktu: 0.71s | Inferensi: 1.00ms

Ringkasan Perbandingan Model:

Model	Subset Acc	Macro F1	Hamming	Train Time	Inference/Sample
SNN	1.0000	1.0000	0.0000	27.14	4.095 ms
RF	0.9000	0.9333	0.0200	23.68	5.291 ms
SVM	1.0000	1.0000	0.0000	0.71	0.999 ms

Laporan Detail per Gas (Random Forest):

	precision	recall	f1-score	support
AIR	1.00	1.00	1.00	2
C2H2	1.00	1.00	1.00	2
C2H4	1.00	1.00	1.00	2
H2	1.00	0.50	0.67	2
CH4	1.00	1.00	1.00	2
micro avg	1.00	0.90	0.95	10
macro avg	1.00	0.90	0.93	10
weighted avg	1.00	0.90	0.93	10
samples avg	0.90	0.90	0.90	10

Contoh Output Confidence (Probabilitas) Multi-Label - Random Forest:

Sampel 1:

- Label Asli (Ground Truth) : {'AIR': np.float32(0.0), 'C2H2': np.float32(1.0), 'C2H4': np.float32(0.0), 'H2': np.float32(0.0), 'CH4': np.float32(0.0)}
- Prediksi Confidence Model : {'AIR': '2.5%', 'C2H2': '60.5%', 'C2H4': '3.0%', 'H2': '5.0%', 'CH4': '0.5%'}

Sampel 2:

- Label Asli (Ground Truth) : {'AIR': np.float32(0.0), 'C2H2': np.float32(0.0), 'C2H4': np.float32(0.0), 'H2': np.float32(0.0), 'CH4': np.float32(0.0)}
- Prediksi Confidence Model : {'AIR': '32.5%', 'C2H2': '5.5%', 'C2H4': '0.5%', 'H2': '0.5%', 'CH4': '97.0%'}

[4/8] Running Comprehensive Evaluations...

K-Fold CV (5 folds) Random Forest

Acc: 0.8800 0.1166
F1: 0.8307 0.1080

K-Fold CV (5 folds) SVM

Acc: 0.8400 0.0490
F1: 0.7773 0.0666

LOMO Random Forest

LOMO Subset Acc: 0.8600
LOMO Macro F1: 0.9214

```

LOMO SVM
LOMO Subset Acc: 0.8400
LOMO Macro F1: 0.9109

[5/8] LOCO (Leave-One-Channel-Out) RF...
Tanpa mos0_mV: Acc = 0.8787
Tanpa mos1_mV: Acc = 0.8591
Tanpa mos2_mV: Acc = 0.8591
Tanpa mos3_mV: Acc = 0.8591
Tanpa mos4_mV: Acc = 0.8787
Tanpa mos5_mV: Acc = 0.8787
Tanpa mos6_mV: Acc = 0.8591
Tanpa mos7_mV: Acc = 0.8591
Tanpa mos8_mV: Acc = 0.8395
Tanpa mos9_mV: Acc = 0.8395
Tanpa mos10_mV: Acc = 0.8395
Tanpa mos11_mV: Acc = 0.8591
Tanpa mos12_mV: Acc = 0.8591
Tanpa mos13_mV: Acc = 0.8787
Tanpa mos14_mV: Acc = 0.8395
Tanpa mos15_mV: Acc = 0.8395
Tanpa temp_chamber_C: Acc = 0.8591
Tanpa temp_oil_C: Acc = 0.8395
Tanpa humidity_chamber_pct: Acc = 0.8591
Tanpa humidity_oil_pct: Acc = 0.8591
Tanpa ina226_1_vbus_V: Acc = 0.8395
Tanpa ina226_1_current_mA: Acc = 0.8591
Tanpa ina226_1_power_W: Acc = 0.8382
Tanpa ina226_2_vbus_V: Acc = 0.8382
Tanpa ina226_2_current_mA: Acc = 0.8591
Tanpa ina226_2_power_W: Acc = 0.8382
Tanpa kria_vbus_V: Acc = 0.8186
Tanpa kria_current_mA: Acc = 0.8382
Tanpa kria_power_W: Acc = 0.8382
Tanpa ndir_ratio: Acc = 0.8591
Tanpa ndir_baseline: Acc = 0.8591
Tanpa ndir_response_pct: Acc = 0.8591
Tanpa ndir_absorbance_au: Acc = 0.8591
Tanpa flow_inlet: Acc = 0.8591

[6/8] Compressing & Exporting SNN...
SNN model disimpan: models/snn_gas_composition.json (19187.7 KB)
  Layer 1: 7530128 (sparsity: 8.2%)
  Layer 2: 12864 (sparsity: 1.1%)
  Layer 3: 645 (sparsity: 0.9%)

[7/8] Exporting RF & SVM...
Random Forest models/rf_model.pkl
SVM models/svm_model.pkl

[8/8] Prediksi 1 sample minyak trafo...

Prediksi Unseen: data/unseen/DGA_PFA1_Baseline_2026-09-23T15-41-33.csv

```

HASIL ANALISIS GAS Virtual GC

AIR	99.5%
C2H2	0.0%
C2H4	0.0%
H2	0.1%
CH4	0.0%

IEC 60599: Normal No Fault
Severity: Normal
Tidak ada fault terdeteksi